

GhostBusters

AI Code Bloat Janitor

TEAM

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THE PROBLEM

The Hidden Cost of AI Coding



The Hook

AI assistants like Copilot and Bob write code 10x faster — but they also leave behind a massive trail of semantic tech debt.



The Pain Point

When developers use AI to refactor or delete features, it often leaves orphaned helper functions, unused dependencies, and dead config flags.



The Impact

Standard linters miss this entirely. Over time, codebases become bloated, confusing, and expensive to maintain.

THE SOLUTION

GhostBusters

An enterprise-grade technical debt orchestrator that finds and liquidates "Ghost Dependencies."



THE MRI MACHINE

GhostBusters

Scans the entire codebase in milliseconds to find the exact coordinates of the dead code.



THE SURGEON

IBM Bob

Receives the coordinates and surgically removes the debt across multiple files, safely.

KEY FEATURES

The "Wow" Factor



AST Semantic Usage Engine

Parses the Abstract Syntax Tree locally to find unused logic that standard regex and linters miss.



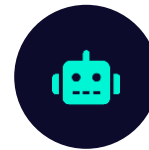
Confidence & Risk Engine

Mathematically scores technical debt and calculates the "Blast Radius" of every change.



Interactive Cleanup Dashboard

A glassmorphism UI where developers can review flagged debt before anything is touched.



Agentic Liquidation

One-click integration with IBM Bob's Subagents to automatically execute multi-file cleanups.

The "50-Credit" Token Strategy



THE CHALLENGE

Dumping a 10,000-file repository into an LLM context window is slow, hallucinatory, and incredibly expensive.



OUR INNOVATION

A lightning-fast local parsing engine (Esprima / TS Compiler) finds the dead code first — only final, verified snippets get sent to IBM Bob.



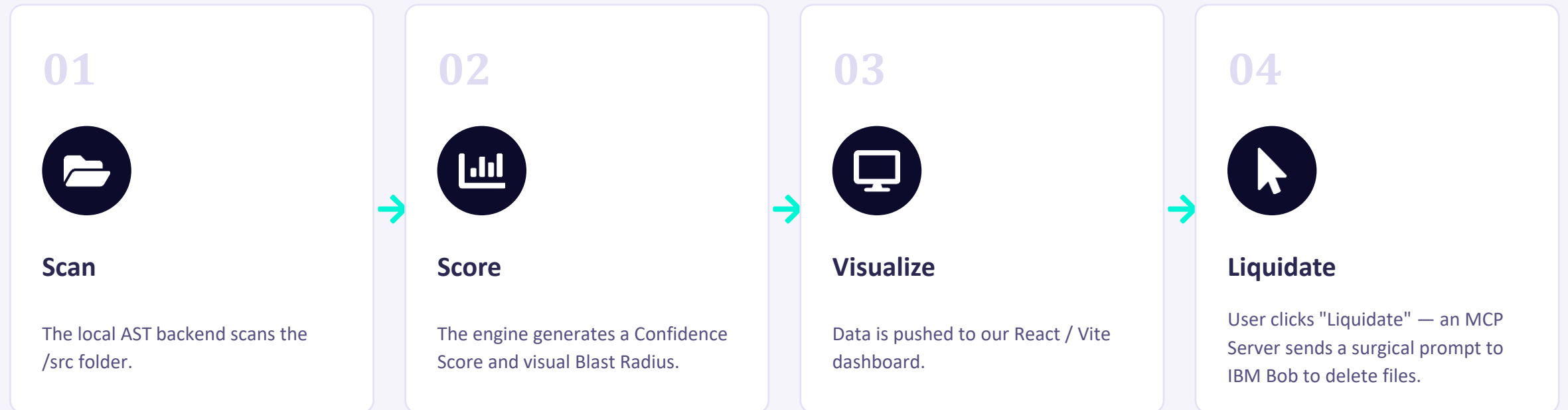
COST EFFICIENCY

Local-First

Token Isolation

The result: IBM Bob becomes highly cost-efficient and load-bearing at enterprise scale.

Architecture



Why It Matters



Saves Money

Reduces LLM token costs by isolating code locally before prompting.



Saves Time

Automates the most tedious part of software engineering — cleaning up old code.



Reduces Risk

Visualizing the blast radius ensures developers don't accidentally break the build.

V2 Vision



Temporal Git Tracking

Connect to Git history to answer "Why does this exist, and who wrote it?" (Mocked in our current UI to show the vision.)



CI/CD Integration

Automatically run GhostBusters as a GitHub Action on every Pull Request to block new code bloat.

THANK YOU

Questions?



GitHub Repo

github.com/priyanshibhatt17/bob-ai-hackathon-GhostBusters

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